1. This question consists of 6 sub-questions. Each item can be answered in a few words or a short paragraph with use of a diagram and/or expression where helpful. Please answer all sub-questions and simplify expressions where possible stating any assumptions made.

a) Compare the key features (and limitations) between an *integrated circuit* and *printed circuit board* based analogue circuit implementation.

[5]

b) Describe the *common base* amplifier using a schematic and comment on the relative levels of its input and output impedances. Describe and justify whether the common base amplifier is better described as a transimpedance amplifier or a unity-gain current amplifier?

[5]

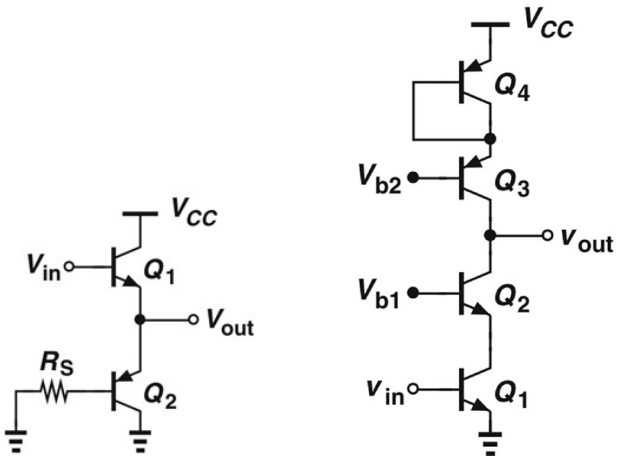
c) Explain how *Miller's theorem* can be exploited in op-amp design in improving *phase margin*.

[5]

d) Describe how a transistor can be configured to approximate the characteristics of a passive inductor (i.e. an *active inductor*). [5]

e) Determine the voltage gain of each circuit shown below in Fig. 1.1. Assume $g_m r_O >> 1$.

[10]



(i) (ii)

Fig. 1.1.

f) Determine the differential and common mode voltage gain of the differential amplifier shown below in Fig. 1.2.

[10]

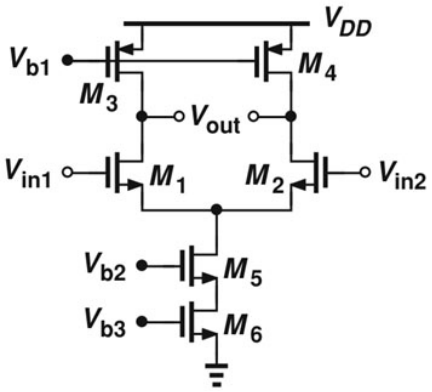


Fig. 1.2.

2. The circuit shown below in Fig. 2.1 is a CMOS buffer consisting of a cascade of two inverter stages with device sizes given in Table 2.1.

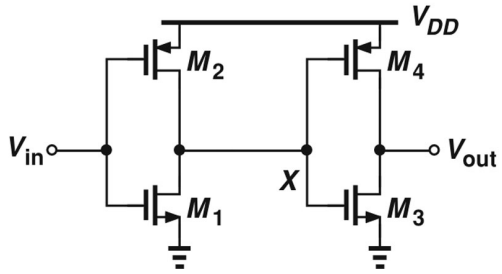


Fig. 2.1.

Table 2.1. Transistor sizes

M_1, M_3	M_2, M_4
5/0.5	10/0.5

Assume all devices are in saturation and assume $\lambda \neq 0$. Use the following expression for the large signal drain current of a MOSFET (in saturation) with corresponding transistor parameters:

$$I_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}\left(V_{GS} - V_{TH}\right)^2$$

$\mu_n C_{ox} = 200\mu A/V^2$ $\quad V_{THN} = 0.4V$ $\quad \lambda_N = 0.1V^{-1}$
$\mu_p C_{ox} = 100\mu A/V^2$ $\quad V_{THP} = -0.5V$ $\quad \lambda_P = 0.2V^{-1}$

a) Compare the operation of the amplifier to a cascade of two common source stages and give two advantages and disadvantages compared to its single transistor counterpart. [5]

b) Determine an expression for the *small signal voltage gain* of the amplifier. Assume all devices are in saturation and assume $\lambda \neq 0$. [5]

c) For a supply voltage of V_{DD}=1.8V, calculate the voltage gain using your answer to part (b). Hint: you will first need to calculate the DC operating point. [5]

Determine expressions for the two pole frequencies associated with nodes X and Vout stating any assumptions made and evaluate. [10]

Where required, use the following parasitic capacitance values: $C_{GS} = (2/3)WLC_{ox}$, $C_{ox} = 20fF/\mu m^2$, $C_{GD} = C_0W$, $C_0 = 0.5fF/\mu m$, $C_{DB} = C_{SB} = 0$.

d) Describe how the disadvantages (or limitations) listed above in part (a) can be addressed. [5]

3. The circuit below in Fig. 3.1 illustrates a high gain microphone preamplifier based on a cascade of two non-inverting op-amp stages. The specifications for the op-amp are given in Table 3.1.

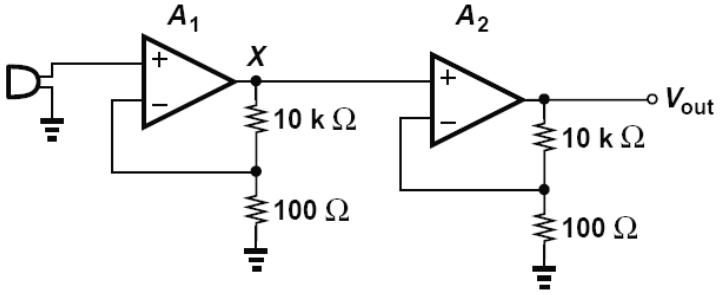


Fig. 3.1

Table 3.1. Op-amp specifications

Parameter	Value
Power supply	3V to 15V
Current consumption	5mA
Input offset voltage	1mV
Common-mode input range	0.5V to VDD-0.2V
Common-mode output range	0.5V to VDD-0.2V
CMRR	70dB
PSRR	70dB
Open loop gain	60dB
Bandwidth (-3dB)	100kHz
Unity gain bandwidth	2MHz
Slew rate	100kV/s
Input impedance	50MΩ
Output impedance	65Ω

a) Discuss any potential challenges in practically implementing (constructing and testing) this circuit due to the op-amp non-idealities based on the provided specifications.

[5]

b) Discuss whether the amplifier is likely to based on CMOS or Bipolar technology? Would you expect the implementation to utilise a N- or P-type (NMOS/PMOS or NPN/PNP) differential pair. Why?

[5]

c) Determine an expression for the closed-loop output impedance and evaluate.

[5]

d) Determine an expression for the *gain error* and evaluate.

[5]

e) In the circuit design of an op-amp, explain what typically limits the: (i) *-3dB bandwidth*; (ii) *slew rate*; (iii) CMRR and PSRR.

[5]

f) Describe two ways in which power consumption can be reduced in op-amp design.

[5]